QC analysis of raw sequencing data

Meryem Stroosma

2025-12-04

Content

Introduction

In this RMarkdown the FastQC and multiQC analysis of the .fastq-files from GSE138852 is described. These .fastq-files contain the raw sequencing data and the phred-scores generated on the Illumina NextSeq 500 high-output lane with the 10x Genomics Chromium Single Cell 3′ Library & Gel Bead Kit v2. This markdown explains how analysis should be done and what the conclusion and discussion is in regards to the research question.

Research question

For this project the main research question is *"How can a reliable and reproducible analysis pipeline be established to identify cell-type-specific differential gene expression from snRNA-seq data in Alzheimer's disease?"*, the first step in answering this question is looking at the quality of sequencing data. The sub-question that is answered in this RMarkdown is as followed: Do the sequencing reads have a high enough Q-score to further analyse the data ($Q \geq 30$)?

Packages and Requirements

In order to run the scripts for QC-analysis a few packages need to be installed first, for analysis an environment was created in miniconda with the command line prompt;

```
conda create -n snrnaseq
```

The conda environment was activated with the prompt

```
conda activate snrnaseq
```

Install the required packages in the conda environment. For obtaining the fastq-files and analyses the following packages are required:

- `sra-tools`
- `fastqc`
- `multiqc`

Run the following commands in the terminal to install the packages:

```
conda install bioconda::sra-tools
conda install bioconda::fastqc
conda install bioconda::multiqc
```

Obtaining the data

The first step in the data-analysis pipeline is obtaining the data. The accession numbers of the GSE138852 dataset that need to be downloaded are listed in the text file *SRR_Acc_List.txt*. To download all the fastq files a bash script is written (`01_download_SRR_fastq.sh`), this script uses the `sra-tools`-package. The first section of this script defines the location were the *SRR_Acc_List.txt* is stored and an output directory, if this is non existing the script will create the output directory prior to fetching the files. After creating an output directory the script uses the *SRR_Acc_List.txt* and reads the first accession number that starts with SRR, it downloads the accession number with fastq-dump from the GEO database and stores it in a compressed format (gzip) to the output directory. After downloading a file it searches for the next accession number in the textfile and repeats the cycle.

The script takes a long time since the downloads are quite large, therefore it is recommended to add `nohup` and & to the command line to run the script on the background of the terminal. Navigate to the scripts folder and run this script from the terminal with the following command:

```
nohup bash 01_download_SRR_fastq.sh &
```

Analysis with FastQC

For FastQC analysis the downloaded fastq-files are used as input data. The fastq files contain the sequencing reads and the quality score (Q-score or Phred score) that are associated with those reads. For each read there are four lines:

1. The first line is the identifier, this is the ID that is generated by the sequencer.
2. The second line contains the nucleotide sequence.
3. The third line is the symbol "+", which separates the sequence from the Q-score
4. The fourth line contains the Q-score this is given as a character from the ASCII table.

Analysis of the fastq files is done using a script (`02_fastqc.sh`) that analyzes all fastq files one after the other. This script uses the `fastqc`-package to create a report, the output of this script are two files:

- A html report, which contains a variety of plots such as the per base sequence quality and per sequence quality scores.
- A zipfile that contains the raw data for each plot in text format.

To run this script it is recommended to add `nohup` and & to the command line since there are quite a lot of files to analyse. Navigate to the scripts folder and run this script from the terminal with the following command:

```
nohup bash 02_fastqc.sh &
```

The html report will automatically be saved to the output folder that was specified in the script, two examples of the plots that were analysed were visualized in figure 1 and figure 2.

In figure 1 an example is shown of the plot that is generated for the per base quality score. On the x-axis the position of the base in the read and on the y-axis the quality score is shown. The colours on the background of the plot are an indication of the quality, with green being a very reliable quality, orange is of less reliable quality and red being of poor quality. The blue line in this plot is the mean quality for each position in the read. In this figure we can also see that a boxplot is generated, with the yellow boxes being the interquartile range (25th to 75th percentile) for that position in the read, and the outer black lines indicate the 10th and 90th percentile.

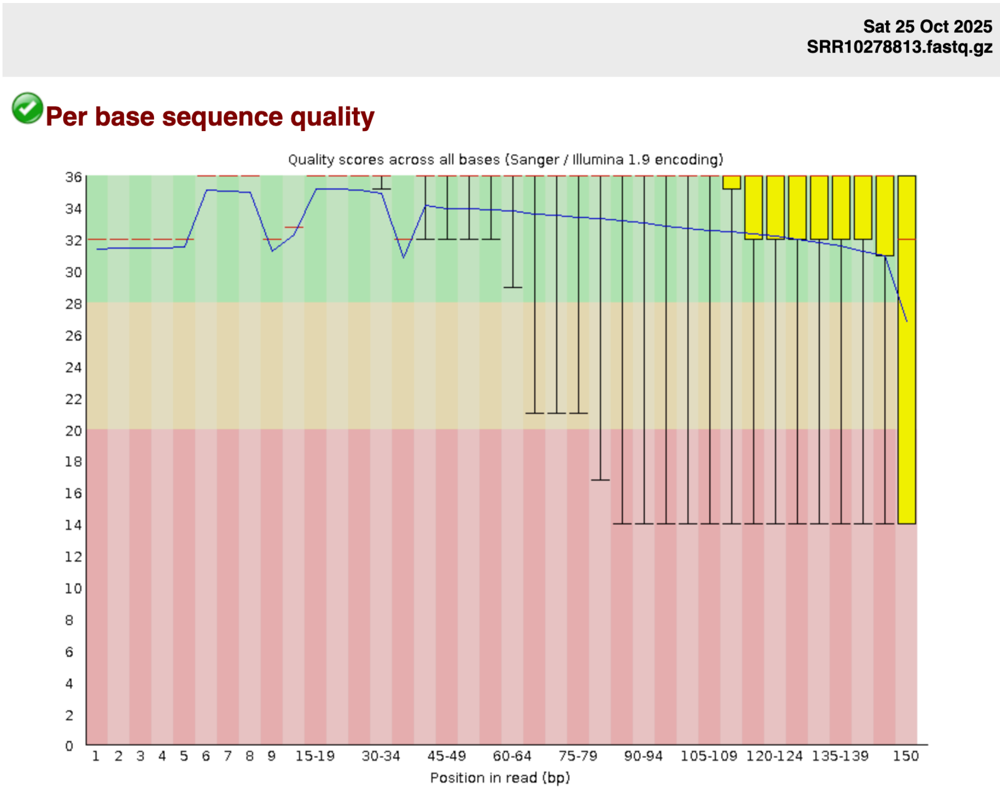

Figure 1: Per base quality score. This figure shows the per base quality score of the data of accession number SRR10278813. This plot was generated with the fast QC package.

In figure 2 an example is shown of the plot that is generated for the per sequence quality score. On the x-axis the mean quality score is shown and on the y-axis the number of sequences with that quality score.

If most of the sequences in this plot have a high quality score it usually means that the sequencing run is successful.

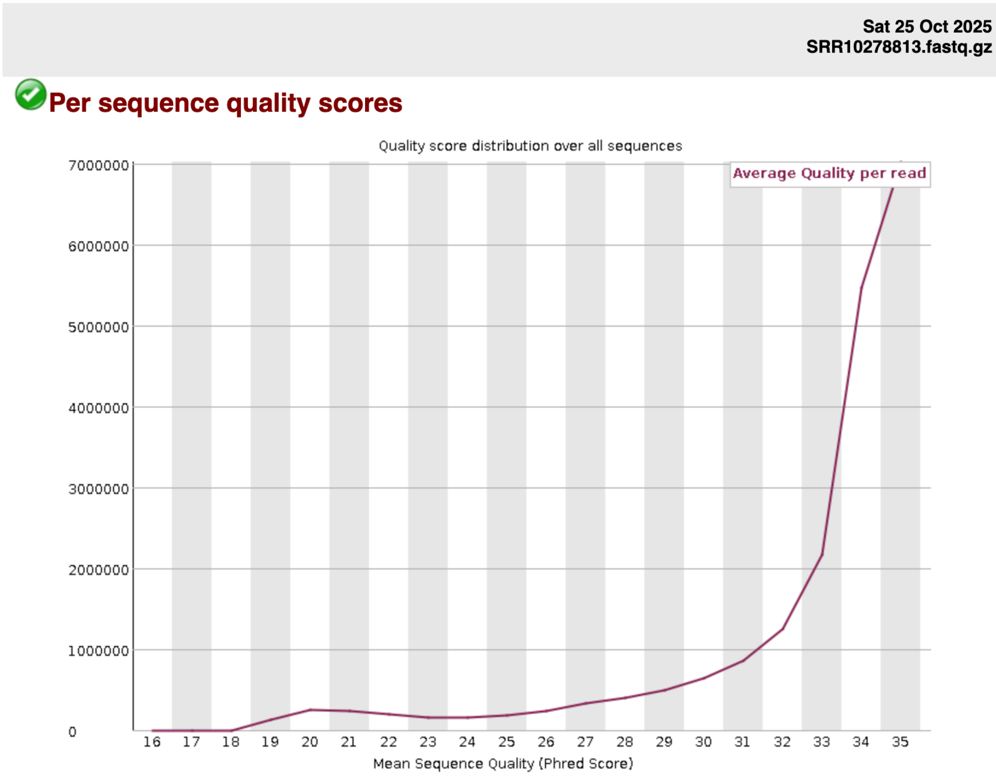

Figure 2: Per sequence quality score. This figure shows the per sequence quality score of the data of accession number SRR10278813. This plot was generated with the fast QC package.

Analysis with MultiQC

The analyses of all the plots that were generated in analysis with FastQC is quite time consuming. To optimize this part of the workflow the package `MultiQC` was used. No script was written for MultiQC, after installing multiQC it is possible to run MultiQC from the command line. In order to run multiQC analysis it is important to first navigate to the directory where the fastQC reports are stored, since multiQC uses these reports as input.

```
cd path/to/fastqc_reports
multiqc .
```

After running multiQC a report is generated in the directory where analysis was performed. This report contains all of the same plots that a fastQC report generates. However a multiQC report contains interactive plots were all of the input reports are layered on top of each other, which makes it easier to compare data of different runs and samples. In Figure 3 an the plot for per base quality score is shown, compared to the fastQC report this one doesn't contain any box plots. Each line represents a different accession number. In the html version of the multiQC report this plot is interactive, if the mouse hovers over one of the lines, the corresponding accession number, position in read and mean phred score are shown.

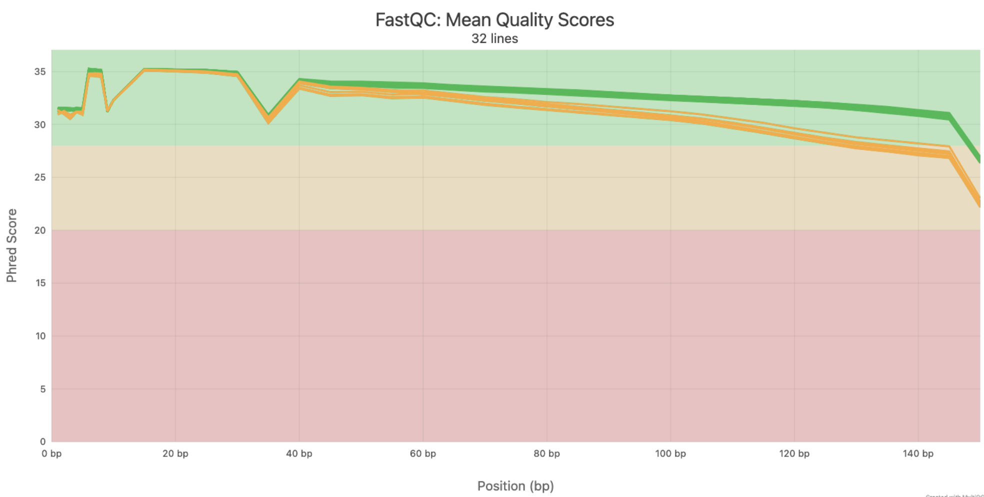

Figure 3: Per base quality score. This figure shows the per base quality score of all of the data from GSE138852. This plot was generated with the multiQC package.